

Campuslands Guatemala Tec Zona 4

DOCUMENTACIÓN

TutorBot – Sistema de Asesorías Académicas



Profesor:

Eduin Enrique Salas Vilorio

Estudiantes:

María José Montepeque Zet

Allison Rocio Vargas Mejía

Guatemala 10 de julio del 2026

DOCUMENTACIÓN

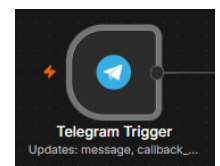
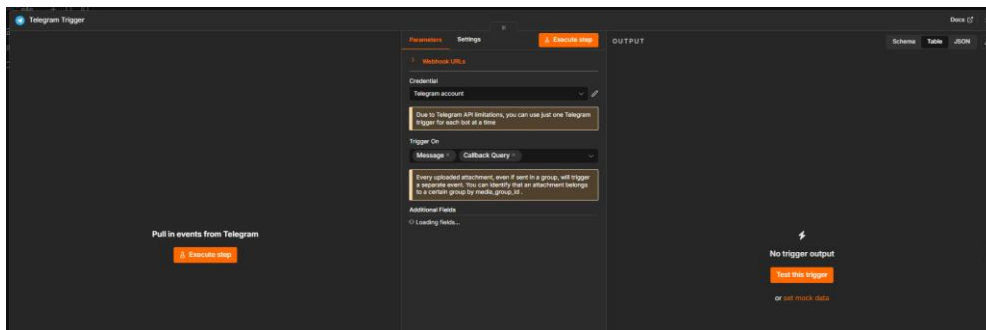
TUTORBOT — SISTEMA DE ASESORIAS ACADÉMICAS

DIAGRAMA FINAL:

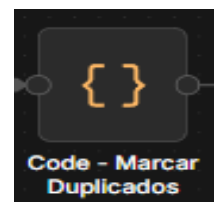
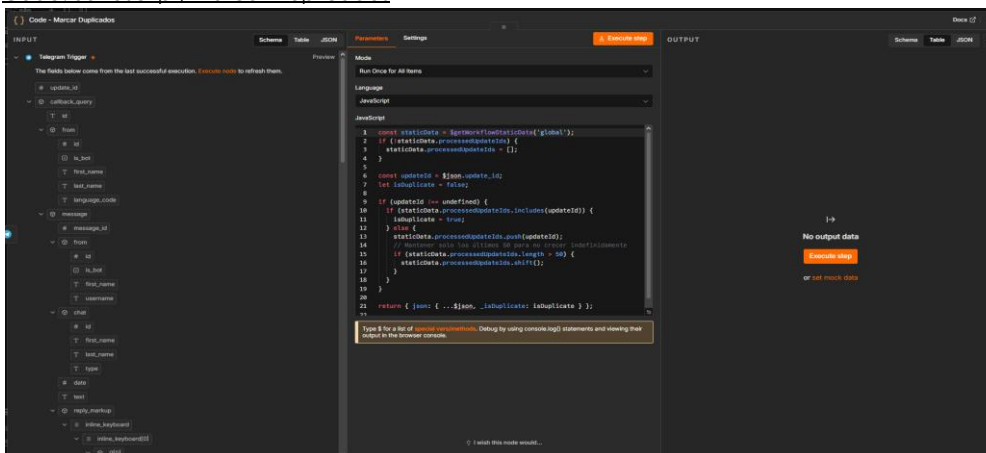


EXPLICACION PASO A PASO:

Telegram Trigger

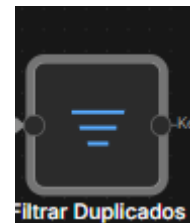


Code: Javascript, Marcar Duplicados



Filtrar Duplicados

The screenshot shows the configuration for the 'Filtrar Duplicados' (Filter Duplicates) node. The 'INPUT' section on the left lists various fields from a previous execution, including 'update_id', 'callback_query', 'from' (with sub-fields 'id', 'is_bot', 'first_name', 'last_name', 'language_code'), 'message' (with 'message_id'), 'chat' (with 'id', 'first_name', 'last_name', 'type'), 'date', 'text', 'reply_markup', 'inline_keyboard', and 'data'. The 'Conditions' section in the center is set to 'False' with the condition '[[\$json._iduplicate]]'. The 'Options' section is currently empty. The 'OUTPUT' section on the right shows 'No output data' and an 'Execute step' button.

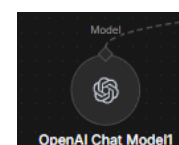


AI Agent



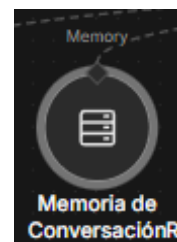
OpenAI Chat Model1

The screenshot shows the configuration for the 'OpenAI Chat Model1' node. The 'INPUT' section on the left lists fields from a previous execution, including 'update_id', 'callback_query', 'from' (with sub-fields 'id', 'is_bot', 'first_name', 'last_name', 'language_code'), 'message' (with 'message_id'), 'chat' (with 'id', 'first_name', 'last_name', 'type'), 'date', 'text', 'reply_markup', 'inline_keyboard', and 'data'. The 'Parameters' section in the center is set to 'OpenAI account' with a dropdown menu. The 'Model' is set to 'gpt-4o-mini'. The 'Use Responses API' checkbox is checked. The 'Built-in Tools' section is currently empty. The 'Options' section is currently empty. The 'OUTPUT' section on the right shows 'No output data' and a note: 'Output will appear here once the parent node is run'.



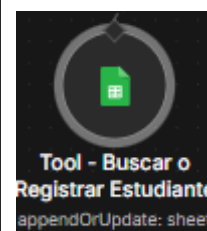
Simple Memory: Memoria de Conversación

The screenshot shows the configuration for the 'Memoria de Conversación' node. The left pane displays the 'INPUT' section with a tree view of fields: 'callback_query' (containing 'id'), 'from' (containing 'id', 'is_bot', 'first_name', 'last_name', 'language_code'), 'message' (containing 'message_id'), 'chat' (containing 'id', 'first_name', 'last_name', 'type', 'date', 'text'), and 'reply_markup' (containing 'inline_keyboard' and 'id'). The right pane shows the 'Parameters' section with a 'Session ID' field set to 'Define below', a 'Key' field containing a JSON path expression, and a 'Context Window Length' field set to '10'. The 'OUTPUT' section on the far right shows 'No output data'.



Google Sheet: Tool - Buscar o Registrar Estudiante

The screenshot shows the configuration for the 'Tool - Buscar o Registrar Estudiante' node. The left pane displays the 'INPUT' section with a tree view of fields: 'callback_query' (containing 'id'), 'from' (containing 'id', 'is_bot', 'first_name', 'last_name', 'language_code'), 'message' (containing 'message_id'), 'chat' (containing 'id', 'first_name', 'last_name', 'type', 'date', 'text'), and 'reply_markup' (containing 'inline_keyboard' and 'id'). The right pane shows the 'Parameters' section with a 'Credential' field, a 'Tool Description' field, a 'Resource' field set to 'Google Sheets account', a 'Sheet Within Document' field, an 'Operation' field set to 'Append or Update Row', a 'Document' field, a 'Sheet' field, a 'Mapping Column Mode' field, a 'Columns to match on' field, and a 'Values to Send' field containing a JSON array. The 'OUTPUT' section on the far right shows 'No output data'.



Google Sheet: Tool – Registrar Sesión

Tool - Registrar Sesión

INPUT Mapping From AI

Filter Duplicates

The fields below come from the last successful execution. [Execute node](#) to refresh them.

callback_query

id

from

id

first_name

last_name

language_code

message

message_id

from

id

first_name

last_name

username

chat

id

first_name

last_name

type

data

test

reply_markup

inline_keyboard

inline_keyboard[0]

id

Parameters

Google Sheets account

Tool Description

Set Manually

Description

Registra o actualiza el estado de sesión del usuario (pantalla_actual, paso_actual, datos, parciales) para el usuario actual del sistema. Requiere: Telegram, wizard.

Resource

Sheet Within Document

Operation

Append or Update Row

Document

By ID

True,Telegram-AbakJuma3DQ

Sheet

By Name

SESSIONS

Mapping Column Mode

Map Each Column Manually

Columns to match on

Telegram_user

The column to use when matching rows in Google Sheets Tool to the input items of this node. Usually an ID.

Values to Send

Telegram_user (using to ...)

[[{"Telegram_user": "chat_id de Telegram"}]]

pantalla_actual

[[{"pantalla_actual": "pantalla de Telegram"}]]

paso_actual

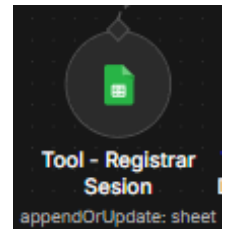
[[{"paso_actual": "pantalla de Telegram"}]]

OUTPUT

This node will automatically retry if it fails

No output data

Output will appear here once the parent node is run



Google Sheet: Tool – Liberar disponibilidad

Tool - Liberar Disponibilidad

INPUT Mapping From AI

Filter Duplicates

The fields below come from the last successful execution. [Execute node](#) to refresh them.

callback_query

id

from

id

first_name

last_name

language_code

message

message_id

from

id

first_name

last_name

username

chat

id

first_name

last_name

type

data

test

reply_markup

inline_keyboard

inline_keyboard[0]

id

Parameters

Google Sheets account

Tool Description

Set Manually

Description

Liberar la disponibilidad de un tutor (vuelve a Libre) cuando se cancela una historia. Parámetro: id_dpto.

Resource

Sheet Within Document

Operation

Update Row

Document

By ID

True,Telegram-AbakJuma3DQ

Sheet

By Name

DISPONIBILIDAD

Mapping Column Mode

Map Each Column Manually

Columns to match on

id_dpto

The column to use when matching rows in Google Sheets Tool to the input items of this node. Usually an ID.

Values to Update

id_dpto (using to match)

[[{"id_dpto": "id_dpto de la fila a liberar"}]]

estado

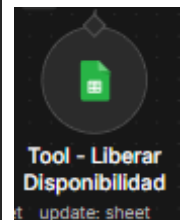
Libre

OUTPUT

This node will automatically retry if it fails

No output data

Output will appear here once the parent node is run



Google Sheet: Tool – Cancelar Tutoria

Tool - Cancelar Tutoria

INPUT Mapping From AI

Filter Duplicates

The fields below come from the last successful execution. [Execute node](#) to refresh them.

update_id

callback_query

from

id

is_bot

first_name

last_name

language_code

message

message_id

chat

chat_id

text

text_name

type

data

text

reply_markup

inline_keyboard

data

Parameters Settings

Credential

Google Sheets account

Tool Description

Set Manually

Description

Cancelar una tutoria existente: pone su estado en 'Cancelar'. Parametro: id_tutoria.

Resource

Sheet Writer Document

Operation

Update Row

Document

By ID

TUTORIAS

Sheet

By Name

TUTORIAS

Mapping Column Mode

Map Each Column Manually

Columns to match on

id_tutoria

The column to use when matching rows in Google Sheets Tool to the input items of this node. Usually an ID.

Values to Update

id_tutoria (existing match)

[[{"id_tutoria": "id_tutoria", "estado": "Cancelar"}]]

estado

Cancelar

Add column to send

Options

OUTPUT

This node will automatically retry if it fails

No output data

Output will appear here once the parent node is run

Tool - Cancelar Tutoria

update: sheet

Google Sheet: Tool – Consultar Estado

Tool - Consultar Estado

INPUT Mapping From AI

Filter Duplicates

The fields below come from the last successful execution. [Execute node](#) to refresh them.

update_id

callback_query

from

id

is_bot

first_name

last_name

language_code

message

message_id

chat

chat_id

text

text_name

type

data

text

reply_markup

inline_keyboard

data

Parameters Settings

Credential

Google Sheets account

Tool Description

Set Manually

Description

Consulta el estado de todas las tutorias de un estudiante, incluyendo su id, para asociarlo (id) para obtener el estado de la tutoria.

Resource

Sheet Writer Document

Operation

Get Row(s)

Document

By ID

TUTORIAS

Sheet

By Name

TUTORIAS

Filters

Column

id_estudiante

Value

[[{"id_estudiante": "id_estudiante", "id_tutoria": "id_tutoria"}]]

Add Filter

Combine Filters

AND

Options

No properties

OUTPUT

This node will automatically retry if it fails

No output data

Output will appear here once the parent node is run

Tool - Consultar Estado

read: sheet

Google Sheet: Tool – Bloquear Disponibilidad

Tool - Bloquear Disponibilidad

INPUT Mapping From AI

The fields below come from the last successful execution. [Execute node](#) to refresh them.

callback_query

from

message

chat

reply_markup

Parameters Settings

Google Sheets account

Tool Description

Set Manually

Description

Actualiza la fila de DISPONIBILIDAD del tutor asignado, manteniéndola como 'Ocupado' para evitar doble reservación. (Reservación de última hora de hoy a hoy).

Resource

Sheet Within Document

Operation

Update Row

Document

By ID

Sheet

By Name

DISPONIBILIDAD

Mapping Column Mode

Map Each Column Manually

Columns to match on

Values to Update

Use the 'Minimize API Calls' option for greater efficiency if your sheet is uniformly formatted without gaps between columns or rows

No output data

Output will appear here once the parent node is run

Tool - Bloquear Disponibilidad update: sheet

Google Sheet: Tool – Registrar Tutoria

Tool - Registrar Tutoria

INPUT Mapping From AI

The fields below come from the last successful execution. [Execute node](#) to refresh them.

callback_query

from

message

chat

reply_markup

Parameters Settings

Google Sheets account

Tool Description

Set Manually

Description

Registra una nueva tutoría en la hoja TUTORIAS con estado 'Asignado' tras la confirmación del estudiante. (Si se le solicita que registre automáticamente, así lo hará).

Resource

Sheet Within Document

Operation

Append Row

Document

By ID

Sheet

By Name

TUTORIAS

Mapping Column Mode

Map Each Column Manually

Columns to match on

Values to Send

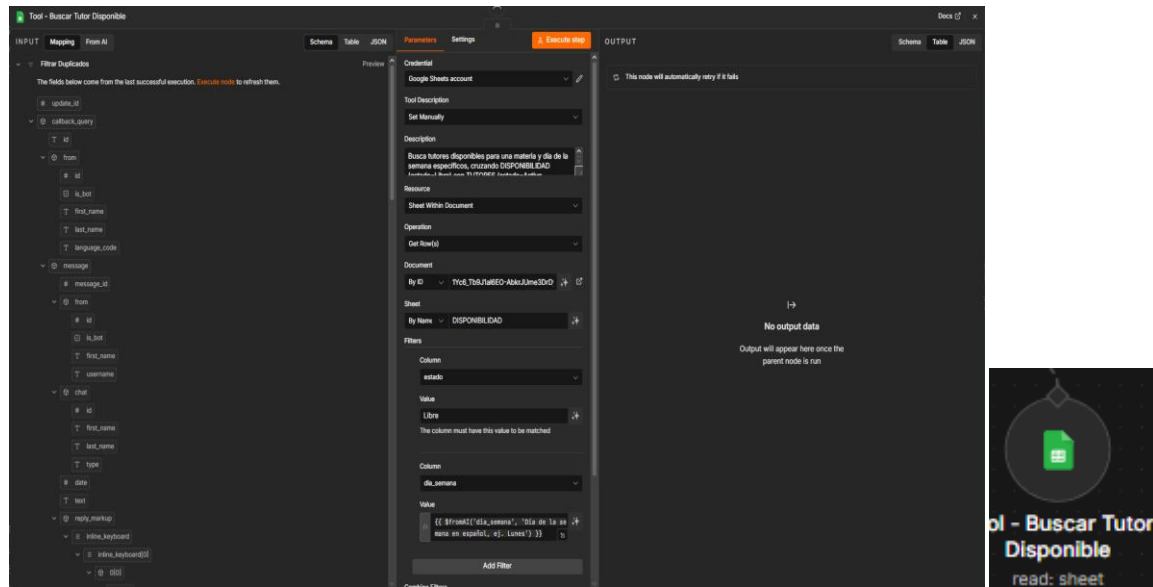
Use the 'Minimize API Calls' option for greater efficiency if your sheet is uniformly formatted without gaps between columns or rows

No output data

Output will appear here once the parent node is run

Tool - Registrar Tutoria append: sheet

Google Sheet: Tool – Tutor Disponible



Tool - Buscar Tutor Disponible

INPUT Mapping From AI

Filter Duplicates

The fields below come from the last successful execution. [Execute node](#) to refresh them.

id

first_name

last_name

language_code

message_id

chat

date

reply_markup

inline_keyboard

data

Parameters Settings

Google Sheets account

Tool Description

Set Manually

Description

Resource

Sheet Within Document

Operation

Get Row(s)

Document

By ID

Sheet

By Name

Filters

Column

Value

Libre

The column must have this value to be matched

Column

Value

Add Filter

OUTPUT

This node will automatically retry if it fails

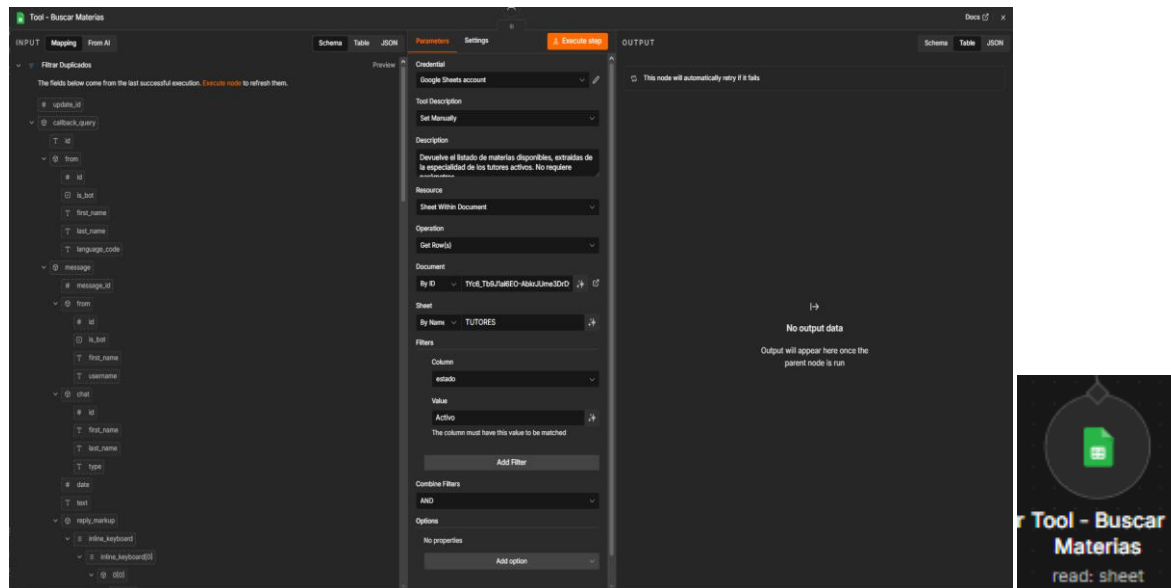
No output data

Output will appear here once the parent node is run

Tool - Buscar Tutor Disponible

read: sheet

Google Sheet: Tool – Buscar Materias



Tool - Buscar Materias

INPUT Mapping From AI

Filter Duplicates

The fields below come from the last successful execution. [Execute node](#) to refresh them.

id

first_name

last_name

language_code

message_id

chat

date

reply_markup

inline_keyboard

data

Parameters Settings

Google Sheets account

Tool Description

Set Manually

Description

Resource

Sheet Within Document

Operation

Get Row(s)

Document

By ID

Sheet

By Name

Filters

Column

Value

Activo

The column must have this value to be matched

Combine Filters

AND

Options

No properties

Add option

OUTPUT

This node will automatically retry if it fails

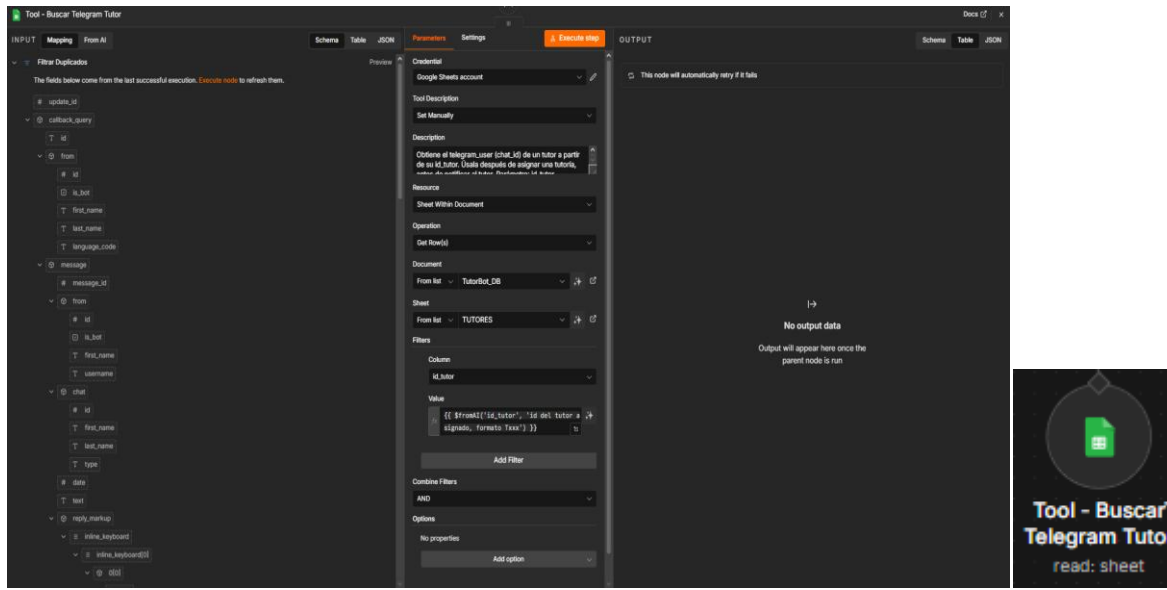
No output data

Output will appear here once the parent node is run

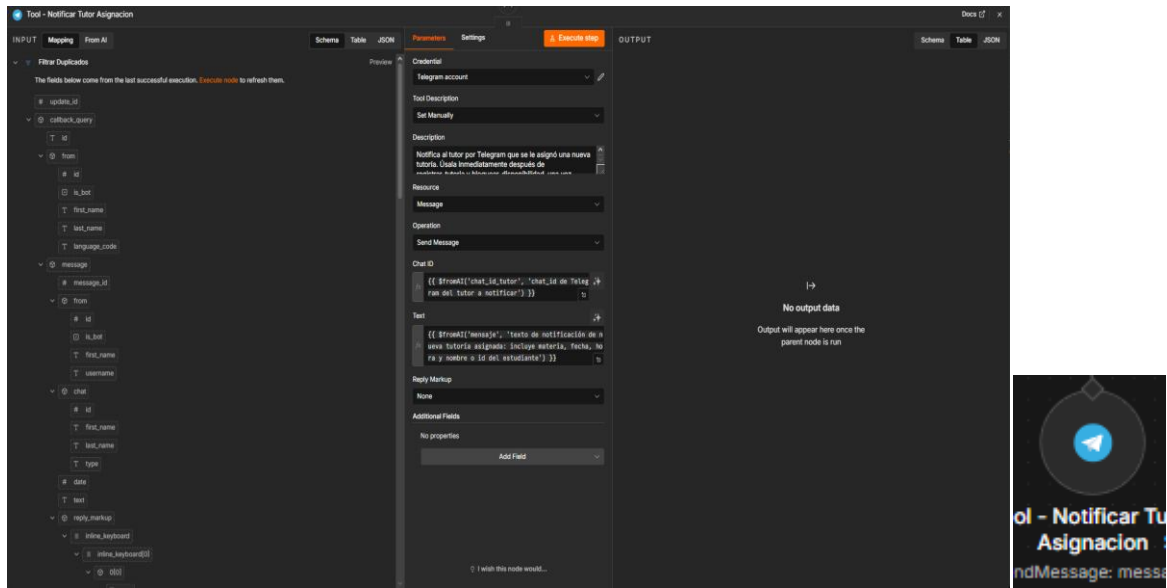
Tool - Buscar Materias

read: sheet

Google Sheet: Tool – Buscar Telegram Tutor



Telegram, Send Message: Tool – Notificar tu asignación



Google Sheet: Tool - Obtener Siguiente ID Tutoria

The screenshot shows the 'Tool - Obtener Siguiente ID Tutoria' interface. The INPUT section on the left lists various fields like 'update_id', 'callback_query', 'from', 'message', 'chat', 'reply_markup', and 'inline_keyboard'. The Mapping section shows a 'From' node with a 'message' object. The Parameters section shows a 'Google Sheets account' dropdown and a 'Tool Description' field. The OUTPUT section on the right shows 'No output data'.

Code Javascript: Parsear Salida IA

The screenshot shows the 'Code - Parsear Salida IA' interface. The INPUT section on the left shows an 'AI Agent' node. The Mapping section shows a 'Code - Parsear Salida IA' node. The Parameters section shows a 'Language' dropdown set to 'JavaScript'. The OUTPUT section on the right shows 'No output data'.

```
1 const raw = $input.json.output;
2 let text = typeof raw === 'string' ? raw : JSON.stringify(raw);
3
4 // Quitar posibles backslashes de markdown ("\" json ... \"")
5 text = text.trim();
6 .replace(/\"\"/g, '');
7 .replace(/\"\"/g, '');
8 .replace(/\"\"/g, '');
9 .trim();
10
11 let parsed = null;
12
13 // Intento 1: parseo directo
14 try {
15   parsed = JSON.parse(text);
16 } catch (e) {
17   // Intento 2: extraer el primer objeto JSON balanceado dentro del
18   // texto
19   const start = text.indexOf('{');
20   const end = text.lastIndexOf('}');
21   if (start >= 0 && end >= 0 && start < end) {
22     try {
23       parsed = JSON.parse(text.slice(start, end + 1));
24     } catch (e2) {
25       parsed = null;
26     }
27   }
28
29   // Fallback: si no se pudo parsear o no tiene la forma esperada,
30   // usamos el texto completo como mensaje conversacional, sin brines,
31   // para que el siguiente GPTWAP pueda dar respuesta en Telegram.
32   if (parsed || typeof parsed !== 'object') {
33     parsed = { message: text, options: {} };
34   }
35 }
```

Code Javascript:

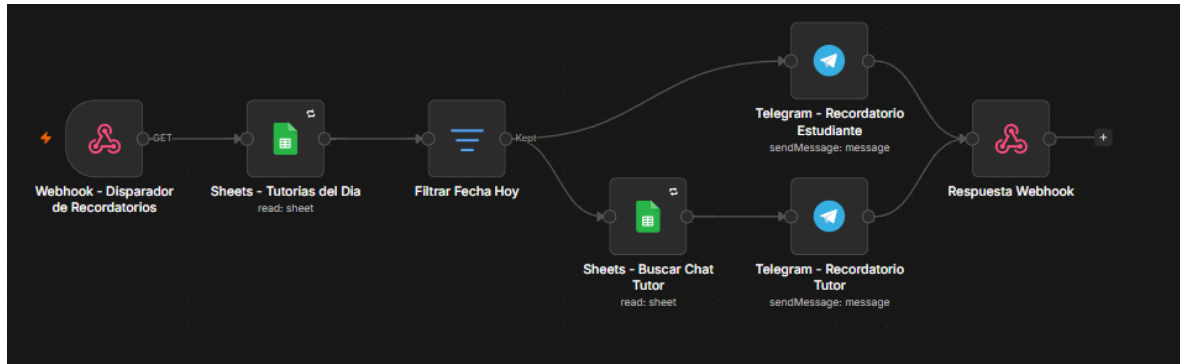
The screenshot shows the Code in JavaScript IDE interface. The main editor displays a JavaScript code snippet for a Telegram chatbot. The code uses the Telegram Bot API to send a message to a chat. The IDE includes a sidebar with project files, a top bar with 'Code in JavaScript' and 'Data' tabs, and a right sidebar with 'OUTPUT' and 'Schema' tabs. A yellow tooltip at the bottom left says 'Type S for a list of special characters/keywords. Debug by using console.log() statements and viewing their output in the browser console.'

```
1 const output = $request.item.json.output;
2 const trigger = $if(telegram.trigger).first().json;
3 const chatId = trigger.message ? trigger.message.chat.id :
4   trigger.callback_query.message.chat.id;
5
6 const body = {
7   chat_id: chatId,
8   parse_mode: "Markdown"
9 };
10
11 if (output.options && output.options.length > 0) {
12   body.reply_markup = {
13     inline_keyboard: output.options.map(o => ({ text: o.text,
14       callback_data: o.value })))
15   };
16 }
17 return { json: { telegrammbody: body } };
```

Post: HTTP Request

The screenshot shows the HTTP Request 1 tool interface. The left sidebar lists several requests, with 'Telegram Trigger' selected. The main panel displays the configuration for this request, including the URL, method (POST), and body (JSON). The right sidebar shows the 'Variables and content' section, which is currently empty.

DIAGRAMA FINAL:



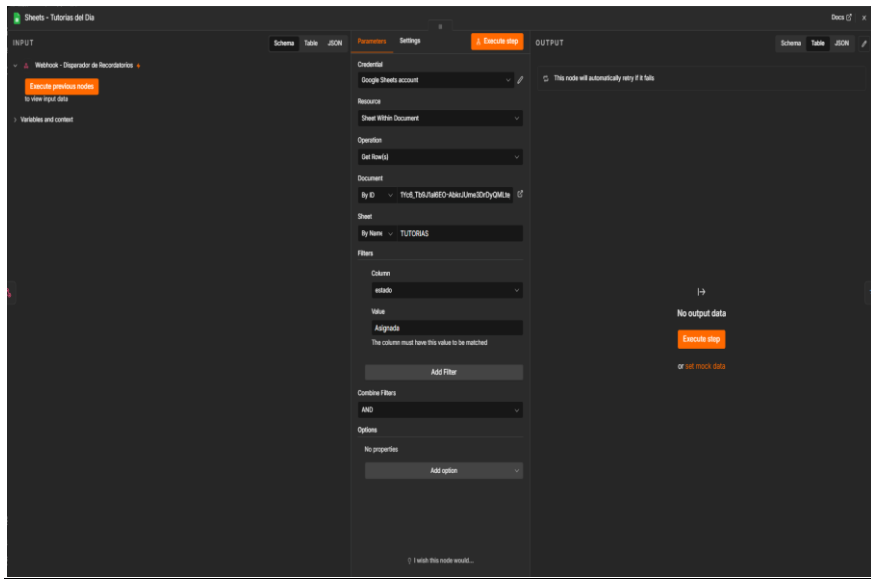
Post: HTTP Request

The screenshot shows the configuration interface for a 'Webhook - Disparador de Recordatorios' node. The 'Parameters' tab is active, displaying the following settings:

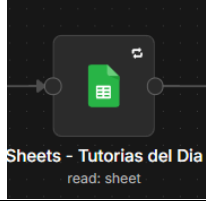
- Webhook URL:** Test URL (Production URL) with the value `https://getWebhook-S675-use.demonstru.ms/webhook-4-est/tutoral-recordatorios`.
- HTTP Method:** GET
- Path:** tutorial-recordatorios
- Authentication:** None
- Response:** Using 'Respond to Webhook' Node
- Options:** No properties

The 'OUTPUT' tab on the right shows 'No trigger output' and a 'Test this trigger' button. Below the main configuration, a small diagram shows the 'Webhook - Disparador de Recordatorios' node with a GET trigger.

Google Sheet: Tool – Tutorías del día

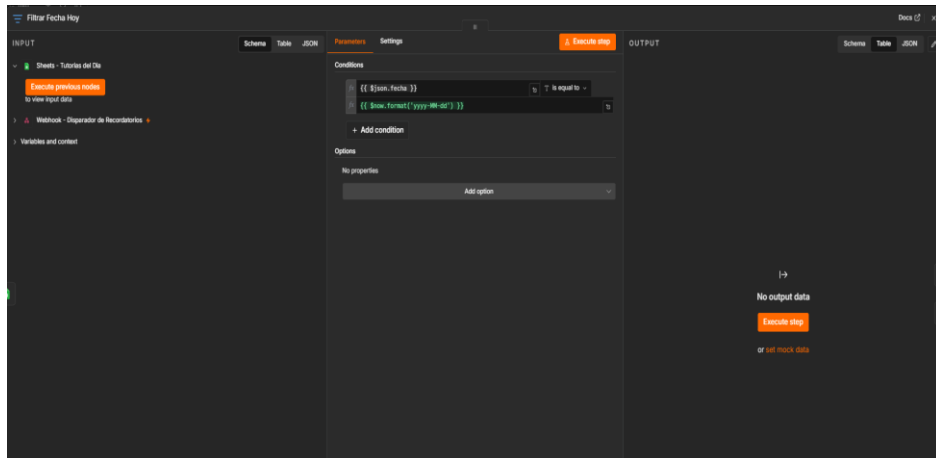


The screenshot shows the n8n workflow editor for a workflow named "Sheets - Tutorías del Día". The "Parameters" tab is selected, displaying configuration for a Google Sheets integration. Key settings include a Google Sheets account, a specific document ID, and a sheet named "TUTORIAS". A filter is applied to the "estado" column, set to "Asignada". The "OUTPUT" panel on the right indicates "No output data".

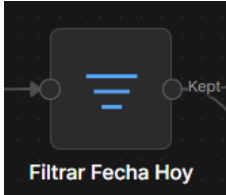


Sheets - Tutorías del Día
read: sheet

Filtrar Fecha Hoy

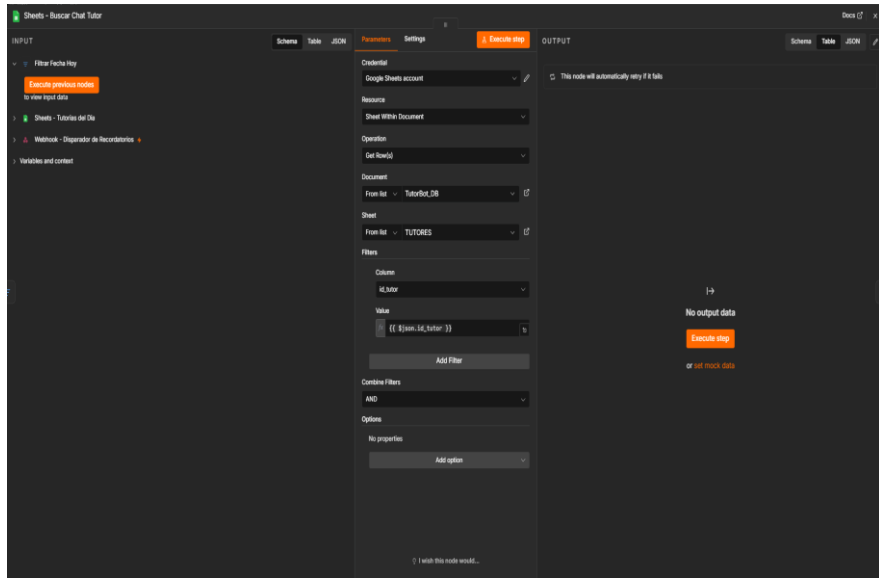


The screenshot shows the n8n workflow editor for a workflow named "Filtrar Fecha Hoy". The "Conditions" tab is selected, displaying a condition where the date is equal to "today". The "OUTPUT" panel on the right indicates "No output data".



Filtrar Fecha Hoy

Google Sheet: Buscar Chat Tutor



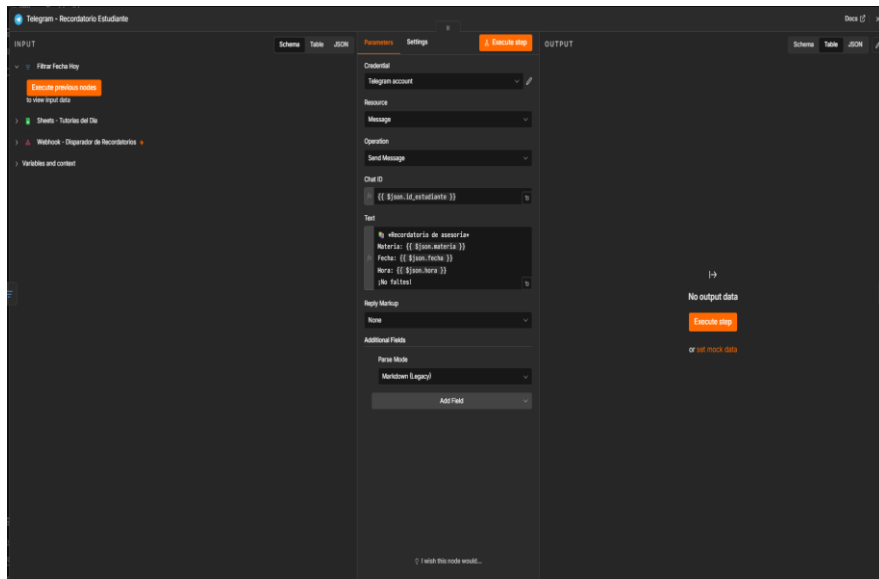
The screenshot shows the n8n interface for a workflow titled "Sheets - Buscar Chat Tutor". The workflow is configured with the following nodes and settings:

- Filter Function**: Set to "Filter Function".
- Google Sheets**: Set to "Read" and "Sheet" "TUTORIAS".
- Telegram**: Set to "Send Message" and "Chat ID" is set to "[[\$json.id_chatTutor]]".

The output panel shows "No output data" and "Execute step" button.

Sheets - Buscar Chat Tutor
read: sheet

Telegram:Recordatorio Estudiante



The screenshot shows the n8n interface for a workflow titled "Telegram - Recordatorio Estudiante". The workflow is configured with the following nodes and settings:

- Telegram**: Set to "Send Message" and "Chat ID" is set to "[[\$json.id_chatTutor]]".
- Telegram**: Set to "Send Message" and "Chat ID" is set to "[[\$json.id_chatTutor]]".
- Telegram**: Set to "Send Message" and "Chat ID" is set to "[[\$json.id_chatTutor]]".

The output panel shows "No output data" and "Execute step" button.

Telegram - Recordatorio Estudiante
sendMessage: message

Telegram: Recordatorio Tutor

The screenshot shows the n8n workflow editor for a Telegram bot named "Recordatorio Tutor". The workflow consists of a single step, "Telegram - Recordatorio Tutor", which is configured with the following parameters:

- Operation:** Send Message
- Chat ID:** {{ \$json.telegram_user_id }}
- Text:** Recordatorio de asessoria
Materia: {{ \$filter[Fecha Hoy].item.json.materia }}
Fecha: {{ \$filter[Fecha Hoy].item.json.fecha }}
Hora: {{ \$filter[Fecha Hoy].item.json.hora }}
- Reply Markup:** None
- Parse Mode:** Markdown (Legacy)

The output panel shows "No output data". A small thumbnail of the workflow is visible in the bottom right corner, labeled "Telegram - Recordatorio Tutor" with the action "sendMessage: message".

Webhook: Respuesta

The screenshot shows the n8n workflow editor for a webhook named "Respuesta Webhook". The workflow consists of a single step, "Webhook - Recordatorio Estudiante", which is configured with the following parameters:

- Response With:** JSON
- Response Body:** {{ \$filter[Fecha Hoy].item.json.materia }}
{{ \$filter[Fecha Hoy].item.json.fecha }}
{{ \$filter[Fecha Hoy].item.json.hora }}

The output panel shows "No output data". A small thumbnail of the workflow is visible in the bottom right corner, labeled "Respuesta Webhook".